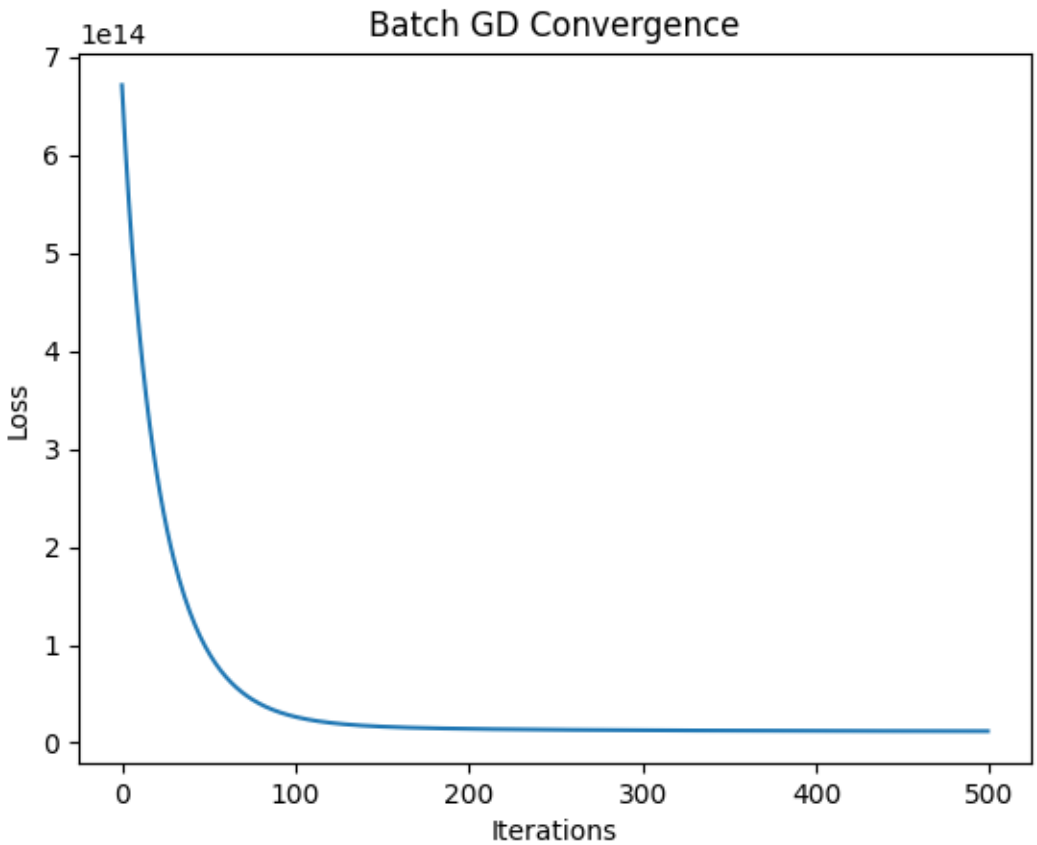
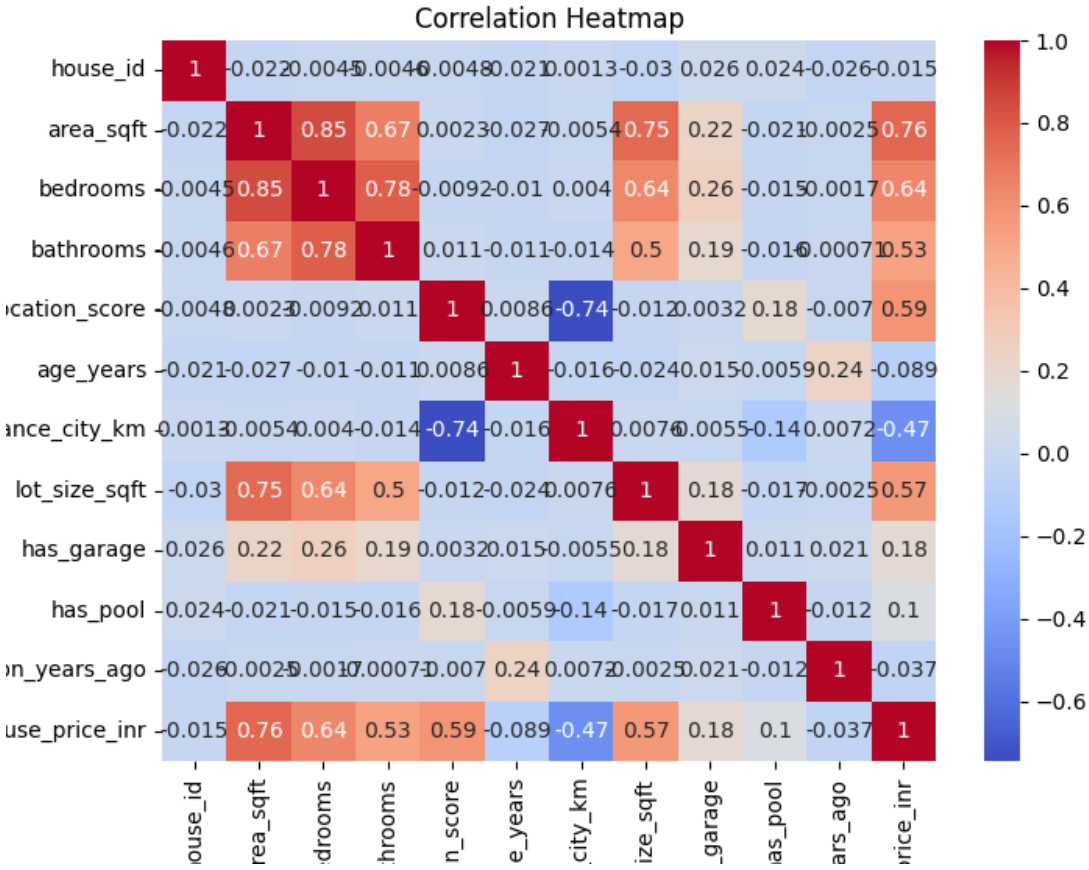
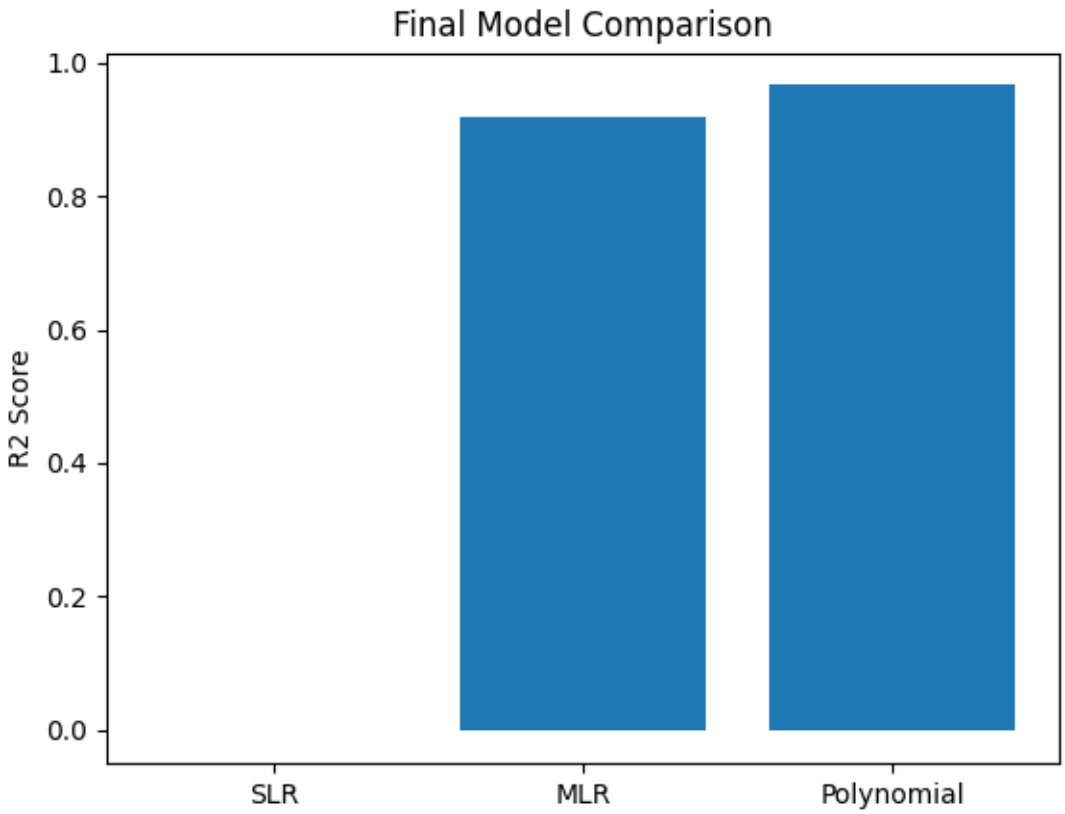
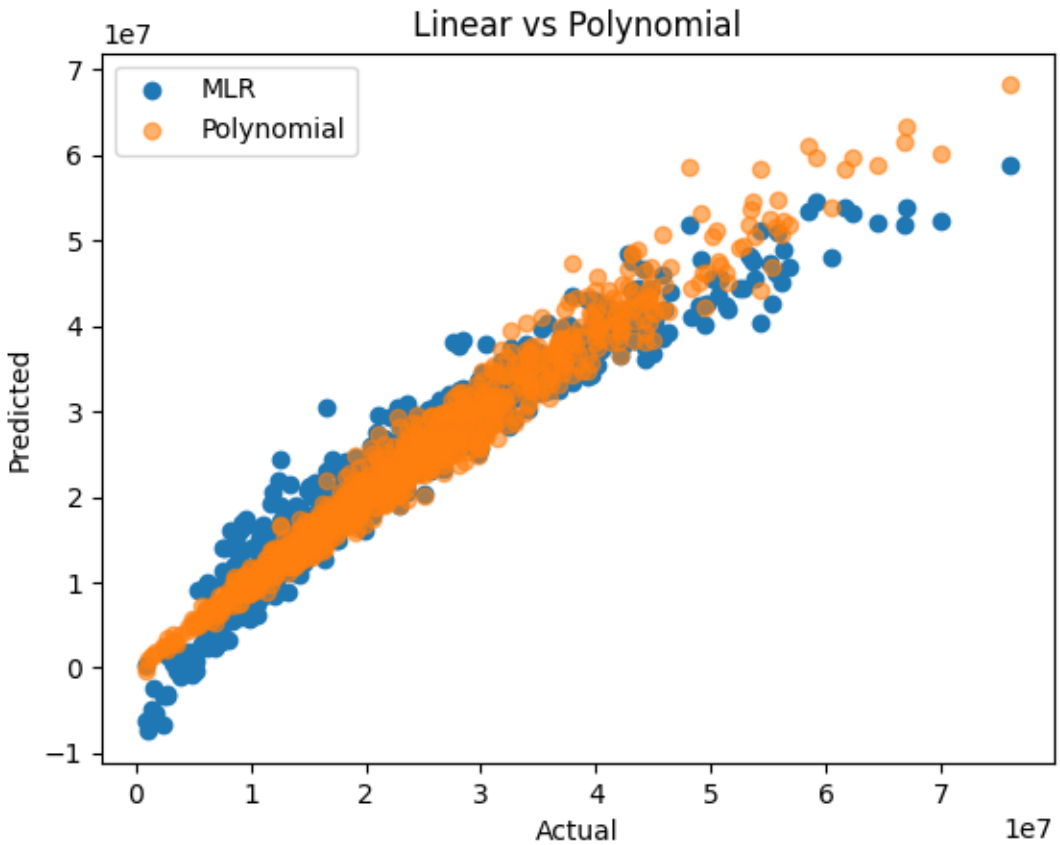
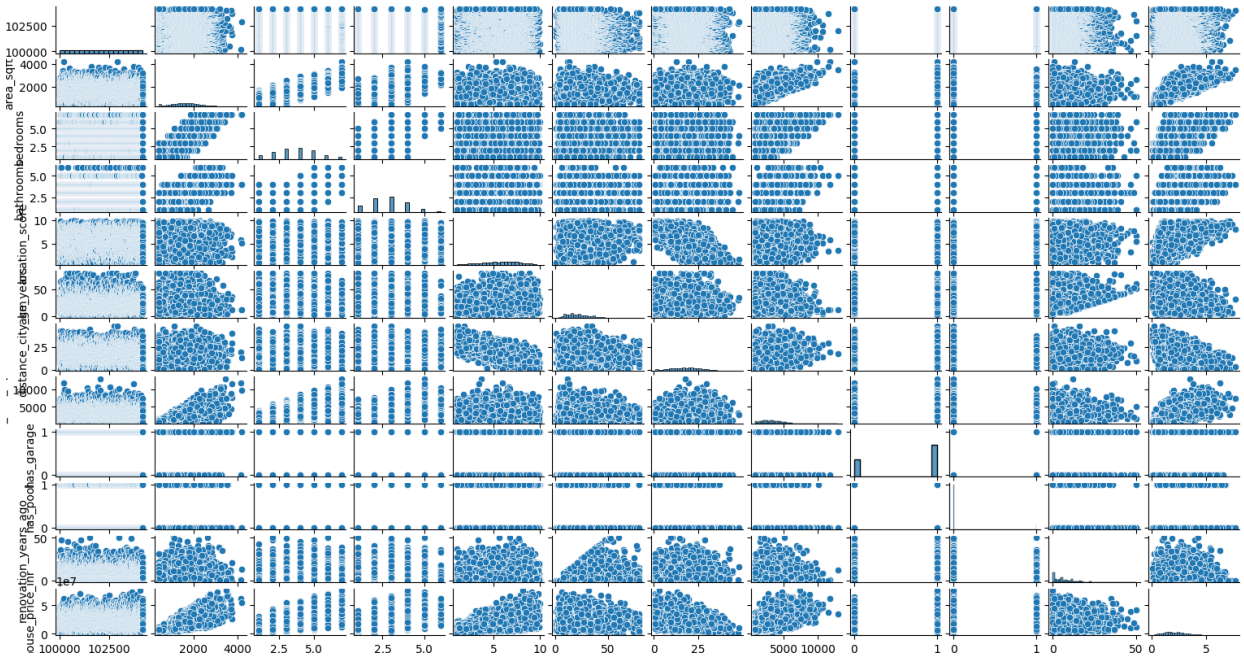

Batch GD Convergence
1e14
7
6
5
4
3
2
1
0
Loss
0
100
200
300
400
500
Iterations


Correlation Heatmap
house_id
area_sqft
bedrooms
bathrooms
cation_score
age_years
ance_city_km
lot_size_sqft
has_garage
has_pool
n_years_ago
use_price_inr
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6


Final Model Comparison
R2 Score
1.0
0.8
0.6
0.4
0.2
0.0
SLR
MLR
Polynomial


Linear vs Polynomial
1e7
MLR
Polynomial
Predicted
Actual
1e7


area_sqft
bedrooms
bathrooms
location_score
renovation_year
age
distance_city_km
has_garage
has_pool
renovation_years_ago
house_price_inr

```
PS T:\RNW Course and Project\Machine Learning\Project 1> & C:/Python314/python.exe "t:/RNW Course and Project/Machine Learning/Project 1/predictive_insight_engi
ne.py"
Columns: Index(['house_id', 'area_sqft', 'bedrooms', 'bathrooms', 'location_score',
       'age_years', 'distance_city_km', 'lot_size_sqft', 'has_garage',
       'has_pool', 'renovation_years_ago', 'house_price_inr'],
      dtype='str')

Independent Variables:
 Index(['house_id', 'area_sqft', 'bedrooms', 'bathrooms', 'location_score',
       'age_years', 'distance_city_km', 'lot_size_sqft', 'has_garage',
       'has_pool', 'renovation_years_ago'],
      dtype='str')

Dependent Variable: house_price_inr

SLR Coefficient: -104.30987233973498
SLR Intercept: 34170263.675823726

SLR Metrics: (153413981238154.6, 9740529.456037272, np.float64(12386039.77218524), -0.001885299389221995, -0.0030808665722641404)

MLR Metrics: (12597222158307.7, 2605648.611505146, np.float64(3549256.5641705445), 0.9177325847899332, 0.9166396601917319)

Polynomial Metrics: (5119996285496.729, 1659159.3088949106, np.float64(2262740.878999787), 0.9665633538093017, 0.9631362074191907)

Batch GD Metrics: (13180848168758.49, 2666529.258188821, np.float64(3630543.784167668), 0.9139211569429229, 0.9127775974337105)

SGD Metrics: (16648737206821.63, 3140205.75955926, np.float64(4080286.412351666), 0.8912737618417235, 0.8898293311415532)

Mini-Batch GD Metrics: (12566709704452.47, 2609414.690968872, np.float64(3544955.529263022), 0.9179318494118349, 0.9168415720489487)

Gradient Descent Comparison:
Batch GD: 3630543.784167668
SGD: 4080286.412351666
Mini-Batch GD: 3544955.529263022

Bias-Variance Indicators:
SLR -> R2: -0.0019, RMSE: 12386039.77
MLR -> R2: 0.9177, RMSE: 3549256.56
Polynomial -> R2: 0.9666, RMSE: 2262740.88
PS T:\RNW Course and Project\Machine Learning\Project 1>
```